

What Do Hidden Markov Models Learn From Financial Returns? Regime Detection, Redundancy, and Why Conditioning Does Not Help

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Abstract

Applied work reports that conditioning a forecaster on a hidden-Markov-model regime improves volatility prediction. We tested that claim under strictly causal filtering, an embargoed train/test boundary, hyperparameters selected inside the training window, and HAR, GARCH(1,1) and EWMA benchmarks. It fails. A pooled ridge on identical features wins in 58 of 60 asset-horizon pairs across 20 assets, and no regime variant achieves a significant improvement in 240 Diebold–Mariano tests.

Redundancy explains the failure. Three trailing-volatility features that the model already reads continuously recover the inferred state with 95.7% accuracy, so gating, shrinkage and feature augmentation each supply estimation noise and no information.

Running the same pipeline on synthetic data whose generating process we control locates the failure in the estimator. Two diagnostics that applied work reports as evidence of regimes fail to identify them. State separation reaches 3.00 in a GARCH world containing no discrete states, against 3.15 under genuine Markov switching. Reported transition persistence comes mostly from including an overlapping rolling-volatility emission: fitted to i.i.d. noise, the model still reports substantial persistence, and dropping that one feature removes it. Because Baum–Welch identifies states by their emission distribution, it recovers states differing in volatility *level*, which trailing volatility already reveals, and misses states differing in volatility *dynamics*, which would carry new information. A de-overlapped diagnostic separates the two worlds and returns -0.001 on SPY, below a GARCH process containing no regimes.

Four ordinary evaluation shortcuts, all four present in our own first implementation, turn a -4% result into a significant $+2\%$ one; correcting the baseline alone flips the sign. We release the pipeline and a checklist.

Keywords

volatility forecasting, regime switching, hidden Markov models, evaluation protocol, data leakage, reproducibility

1 Introduction

Financial time series are not stationary. Volatility clusters, correlations change sign in crises, and the relationship between predictors and returns appears to shift over time. This motivates *regime-switching* models, in which a latent discrete state governs the data-generating process [2, 12]. The natural machine-learning instantiation is a two-stage pipeline: infer a latent state with a hidden Markov model [27], then condition a predictor on that state, either

by hard switching between per-regime experts or by soft probability weighting.

The appeal is obvious, the reported results are positive, and the turn toward machine learning in asset pricing [10] has made such pipelines easy to assemble. We started in the same place. Our first implementation showed a clear improvement over a ridge-regression baseline in forecasting SPY volatility. This paper reports what happened when we tried to make that result robust.

The improvement did not survive. It disappeared once we gave the linear baseline the same care as the proposed model, and it reversed once we corrected the target, the train/test boundary and the model-selection step. We are left with a negative result that decomposes: we can attribute the original effect to individual methodological shortcuts and say how much each one contributed.

Contributions.

- (1) **A causal evaluation, replicated across 20 assets.** We evaluate HMM regime-conditioned volatility forecasting under strict online filtering, an embargoed walk-forward split, nested hyperparameter selection, and a benchmark set that includes HAR [6], GARCH(1,1) [4], and EWMA [17]. Regime conditioning provides no significant gain at any horizon, for any asset, under either MSE or QLIKE [25].
- (2) **A decomposition of the null.** We perturb the training partition and the prediction-time gate independently, and we test the two ways of using the regime signal that avoid gating. The result is an accounting identity: the regime signal costs 1.16 points of RMSE before any gating, gating adds 1.59 more, and correct timing returns 0.69. Each route loses, and we quantify by how much and why.
- (3) **A mechanism.** The three HAR components alone recover the inferred state with 95.7% accuracy. The state re-encodes information the model already holds, and loses precision doing so, so no method of injecting it helps.
- (4) **A simulation study bounding what the estimator can see.** We run the same pipeline on data whose generating process we control. The standard descriptive diagnostics fail to distinguish a world with regimes from one without. Overlapping-window emissions manufacture most of the reported transition persistence, which survives even on i.i.d. noise. And maximising emission likelihood recovers states differing in volatility level while missing states differing in volatility dynamics, so the estimator finds the regimes least worth conditioning on and misses the ones that would help.

- (5) **Equivalence bounds.** We report what size of improvement the data exclude, rather than reporting only a failure to reject. In the median asset-horizon pair, no improvement of any size survives at 95%.
- (6) **Robustness on three further axes.** The null holds under gradient-boosted experts, under a Model Confidence Set that weighs all competitors at once, and under a decision-relevant loss. That last case is the one place where the choice of loss function changes the picture: the regime models become indistinguishable from the pooled ridge instead of trailing it, while the regime-free benchmarks stay significantly ahead of both.
- (7) **A protocol ablation.** Starting from our own first implementation, we measure how much apparent improvement each of four evaluation shortcuts produces, and show that they compound: together they turn a -4% result into a statistically significant $+2\%$ one.
- (8) **Seed sensitivity.** The spread across HMM random initializations alone matches the reported effect size. Applied regime-switching work under-appreciates this.
- (9) **A reproducible pipeline and a checklist** for future work in this area.

Two things we do *not* claim. Market regimes may well exist, and regime-switching models have documented value in asset allocation and risk management [11]. Our claim covers this task, this model class and this benchmark set: the reported forecasting gains come from the evaluation rather than from the method, and the diagnostics we propose cost little enough to run as standard.

2 Related Work

Regime switching. Hamilton’s Markov-switching autoregression [12] established the econometric template. Subsequent work applies regime switching to asset allocation [11] and documents its role in explaining return predictability [2, 28]. Most of this literature is in-sample or uses smoothed (two-sided) state estimates; our concern is the strictly causal, out-of-sample forecasting case.

Volatility forecasting. The benchmark set decides this comparison. GARCH [4, 8] remains hard to beat [13], and the HAR model [6] is the standard reference for realized volatility, whose long-memory behavior it captures with three lagged components. Andersen and Bollerslev [1] traced the apparent poor performance of volatility models to the use of squared daily returns as the evaluation proxy; Patton [25] characterized which loss functions survive such proxy noise, which is why we report both MSE and QLIKE.

Evaluation and leakage. Our methodological concerns are not new. López de Prado [21] introduced purging and embargoing for financial cross-validation. Bailey et al. [3] formalized backtest overfitting, and Harvey et al. [15] argued for higher significance thresholds given the volume of tested strategies. Kaufman et al. [19] give the general taxonomy of leakage into which our embargo and nested-selection corrections fall. Welch and Goyal [29] performed the canonical re-evaluation of equity-premium predictors and found most did not survive out-of-sample. Outside finance, Ferrari Dacrema et al. [9] showed that weak baselines account for most reported progress in neural recommendation, Makridakis et

al. [22] made the analogous point for ML forecasting, and Kapoor and Narayanan [18] document leakage as a cross-disciplinary reproducibility problem. Our contribution is to run this style of audit on the specific, popular case of HMM-based regime-conditioned forecasting, and to attribute the effect to individual protocol choices rather than reporting a null alone.

3 Problem Setup

3.1 Data

We use daily SPY (S&P 500 ETF) OHLCV data from February 2005 to July 2026, obtained from Yahoo Finance and adjusted for splits and dividends. Let p_t be the adjusted close and $r_t = \log p_t - \log p_{t-1}$ the daily log return. After feature construction and the removal of warm-up and trailing rows the sample contains 5,391 trading days, of which 3,771 (February 2011 to February 2026) are used for out-of-sample evaluation.

3.2 Targets

The forecasting target is realized volatility over the next h days,

$$y_t^{(h)} = \text{RV}_{t+1:t+h} = \sqrt{\frac{1}{h} \sum_{i=1}^h r_{t+i}^2}, \quad h \in \{1, 5, 20\}. \quad (1)$$

For $h = 1$ this reduces to $|r_{t+1}|$, so the family nests the single-day absolute-return target.

This definition deserves emphasis because the error it guards against is easy to make and hard to see. A similar-looking definition, $\tilde{y}_t^{(h)} = |r_{t+h}|$, gives the absolute return of a *single* day h steps ahead. It aliases: $\tilde{y}_t^{(1)}$, $\tilde{y}_t^{(5)}$ and $\tilde{y}_t^{(20)}$ are the same series at different lags. One diagnostic exposes it. Their sample standard deviations are 0.009186, 0.009185 and 0.009188, identical to four significant figures, whereas the aggregated targets in Eq. (1) give 0.009186, 0.007508 and 0.006837, declining with h as aggregation requires. Under the aliased definition, varying h varies the lead time of one unchanged and very noisy prediction problem. We report results for both. The aliased variant forms the first stage of the protocol ablation in Section 6.3.

3.3 Features

All features at time t use information available at t : lagged returns $r_{t-1}, \dots, r_{t-20}$; rolling means and standard deviations of returns over 5, 10 and 20 days; log volume change and rolling volume z -scores; and Wilder’s RSI. We additionally include the three HAR components

$$\text{RV}_t^{(w)} = \sqrt{\frac{1}{w} \sum_{i=0}^{w-1} r_{t-i}^2}, \quad w \in \{1, 5, 22\}, \quad (2)$$

so that every model in the comparison, including the regime-conditioned one, has access to the information the HAR benchmark uses. The final feature set has 35 columns.

4 Method

4.1 Causal regime inference

We fit a Gaussian HMM with K states and diagonal covariances on the three-dimensional emission vector $x_t = (r_t, |r_t|, \text{RV}_t^{(20)})$, standardized using training-window statistics only. Parameters

$(\pi, A, \{\mu_k, \Sigma_k\})$ are estimated by Baum–Welch on the training window.

The regime probability used at prediction time is the *filtered* posterior

$$\gamma_t(k) = \Pr(z_t = k \mid x_{1:t}), \quad (3)$$

computed by the forward recursion in log space,

$$\log \alpha_t(k) = \log p(x_t \mid z_t=k) + \log \sum_j \exp(\log \alpha_{t-1}(j) + \log A_{jk}), \quad (4)$$

normalized at each step. Regime-switching pipelines leak here more than anywhere else. The smoothed posterior $\Pr(z_t = k \mid x_{1:T})$ returned by a standard `predict_proba` call, and the Viterbi path returned by `predict`, both condition on the *entire* sequence including the future, so forming a forecast at time t from either one uses data the forecaster cannot have. We implement the forward filter ourselves, and we warm-start the test-window filter from the final training-window belief $\alpha_{T_{\text{train}}}$ instead of resetting it to π , matching what someone running the model in real time would hold.

4.2 Label switching across folds

Baum–Welch state indices are arbitrary: nothing ties “state 0” in one fit to “state 0” in the next. Since we refit the HMM on every walk-forward window, concatenating raw state labels across folds interleaves different meanings. The symptom hides well. Forecasts survive intact, because within a fold we fit and apply the experts consistently, but every per-regime table and regime-shading plot then shows apparent regime changes at fold boundaries that are pure relabelling.

We therefore sort states after each fit by the mean of their volatility emission, making state 0 the calmest and state $K - 1$ the most volatile. This relabels and does nothing else: the likelihood, the filtered posteriors and every forecast stay the same, while regime identities become comparable across folds.

4.3 Regime-conditioned prediction

Given filtered probabilities, we fit one ridge expert per regime. Training rows are assigned to expert k by $\arg \max_k \gamma_t(k)$; regimes with fewer than 200 training rows fall back to a globally fitted expert. Predictions combine the experts either by hard switching, $\hat{y}_t = f_{\arg \max_k \gamma_t(k)}(x_t)$, or by soft weighting, $\hat{y}_t = \sum_k \gamma_t(k) f_k(x_t)$.

Each expert is a standardized ridge whose penalty we choose by generalized cross-validation on its own training subsample. Fairness depends on this. Feature standard deviations in our set span four orders of magnitude, from 2.3×10^{-3} for a 20-day mean return to 11.4 for RSI, so a fixed penalty on unscaled inputs shrinks some coefficients far harder than others. Handicap a baseline that way and it stops testing anything.

4.4 Factorized ablations

A single “shuffle the regimes” control answers two questions at once and separates neither. We therefore perturb the training partition and the prediction-time gate independently, giving the five arms in Table 1.

`GATE_SHUFFLE` carries the most weight. We train the experts on genuine regimes, then hand the model a probability vector drawn from the correct marginal distribution but attached to the wrong

Table 1: Ablation design. “true” uses filtered probabilities; “shuffle” applies a random time permutation, preserving the marginal distribution of regime probabilities but destroying their alignment with the market; “single” puts all mass on one state.

Arm	Train gate	Test gate	Isolates
NORMAL	true	true	full model
GATE_SHUFFLE	true	shuffle	value of regime <i>timing</i>
EXPERT_SHUFFLE	shuffle	true	value of regime <i>experts</i>
BOTH_SHUFFLE	shuffle	shuffle	conventional shuffle control
SINGLE	single	single	pooled ridge (control)

day. If knowing *which regime we are in today* matters, this arm has to degrade.

We fell into one design trap worth naming. An earlier version of this ablation suite included a “uniform gate” arm meant to remove regime information while keeping the mixture structure. Since we assign experts by `arg max`, a uniform probability vector selects state 0 for every row, so that arm collapsed to a single globally fitted expert and returned numbers identical to the pooled-ridge control. It looked like an informative ablation and duplicated one we already had.

5 Evaluation Protocol

Walk-forward splits. We use a rolling origin: six years of training (about 1,500 observations) followed by one year of testing, advanced one year at a time, giving 15 folds spanning February 2011 to February 2026. The HMM and all model parameters are refit on each training window.

Embargo. Since $y_t^{(h)}$ depends on returns through $t + h$, the final h rows of any training window carry targets that extend into the test period. We purge them [21]. Skip this step and at $h = 20$ the model trains on twenty observations whose labels come in part from the data about to evaluate it.

Nested selection. We select the number of regimes $K \in \{2, 3, 4\}$ and the gating rule, hard or soft, *within* each training window, on a trailing 25% validation block that we embargo from the inner training portion. We then refit the selected configuration on the full training window and apply it to the test fold. This separates what the method achieves from the best of six out-of-sample numbers.

Metrics. We pool predictions across folds and report RMSE and QLIKE [25], $\text{QLIKE} = \frac{\sigma^2}{\hat{\sigma}^2} - \log \frac{\sigma^2}{\hat{\sigma}^2} - 1$, evaluated on variances. MSE and QLIKE are the two loss families that remain robust to noise in the volatility proxy, so a ranking that holds under both cannot be a proxy artifact. QLIKE diverges as $\hat{\sigma} \rightarrow 0$; we floor forecasts at the 1st percentile of realized values for reporting only.

Significance. We use the Diebold–Mariano test [7] with a Newey–West [24] long-run variance at lag $h - 1$, since overlapping targets make the loss differential autocorrelated by construction, and the Harvey–Leybourne–Newbold [16] small-sample correction. Because each model is compared against several alternatives we apply a Holm–Bonferroni correction within each target.

Table 2: Pooled out-of-sample performance on realized-volatility targets, 15 walk-forward folds, Feb 2011–Feb 2026, $n=3771$. Lower is better. Best per column in bold. Markers report a Holm-corrected Diebold–Mariano test of the regime model against that baseline at the 5% level: * the regime model is significantly better, † significantly worse. The regime model significantly beats only the trivial baselines, and significantly loses to the pooled ridge. QLIKE diverges for near-zero forecasts, which is why the zero and random-walk rows are orders of magnitude above the rest.

Model	$h = 1$		$h = 5$		$h = 20$	
	RMSE	QLIKE	RMSE	QLIKE	RMSE	QLIKE
Zero	0.010847*	1.39e+04	0.010850*	37.7831	0.010852*	8.3736
Train mean	0.008423*	2.7844	0.007042*	1.2120	0.006389*	0.8798
Random walk (RV)	0.009436*	148.9	0.005395	1.0204	0.005706	0.6026
EWMA / RiskMetrics	0.007263	1.8569	0.005234	0.5626	0.005061	0.5077
HAR	0.007133	1.8414	0.004990	0.5362	0.004967	0.4663
GARCH(1,1)	0.007052	1.7780	0.004884	0.5159	0.004913	0.4725
Ridge (pooled)	0.006968†	2.1185	0.004719	0.5585	0.004867†	0.4642
Regime-conditioned (HMM)	0.007225	4.7700	0.004925	0.5975	0.005035	0.4963

6 Results

6.1 Main comparison

Table 2 reports pooled out-of-sample performance.

A pooled ridge on the same features beats the regime-conditioned model at every horizon, and significantly so at $h = 1$ and $h = 20$ after we correct for multiple comparisons. The regime model never beats HAR, GARCH(1,1) or EWMA under either loss at any conventional level. It beats the zero forecast, the training mean, and at $h = 1$ the random walk in realized volatility. Those are the trivial benchmarks, and they reproduce the original result in miniature: pick a weak enough comparison and the method looks convincing.

The two losses disagree about which benchmark ranks first, and the disagreement helps. GARCH(1,1) wins on QLIKE at $h = 1$ and $h = 5$, matching [13]; the feature-rich ridge wins on RMSE everywhere and on QLIKE at $h = 20$. QLIKE punishes under-prediction hard, favouring the conditional-variance models that hold forecasts away from zero, while RMSE rewards the model fitted to minimize squared error. The regime model loses under *both*, which is what leaves the conclusion independent of the loss we chose.

One implementation check: REGIME_SINGLE, the ablation arm that collapses the mixture to one expert, produces predictions bit-wise identical to the pooled ridge at every horizon, confirming that the mixture machinery reduces correctly. Every other regime arm scores worse.

6.2 Tracing the regime signal

Table 3 applies the design of Section 4.4 and carries the central diagnostic. On SPY, GATE_SHUFFLE hands the model regime probabilities attached to the wrong day and costs nothing in RMSE against NORMAL. At $h = 1$ and $h = 5$ it scores *better*.

Read alone, this would make regime timing worthless. The cross-asset evidence in Section 6.4 overturns that reading, and we report it here to warn against drawing mechanism conclusions from one series: pooled over 60 asset-horizon pairs, correct timing carries a

Table 3: Ablation of the regime signal (realized-volatility targets). Δ is RMSE relative to the SINGLE pooled-ridge control, in percent; negative would mean the arm beats the control. Shuffling the gate in time does not degrade the model: at $h = 1$ and $h = 5$ it improves it. If knowing the current regime mattered, the GATE_SHUFFLE row would have to be worse than NORMAL, and it is not.

Arm	$h = 1$		$h = 5$		$h = 20$	
	RMSE	$\Delta\%$	RMSE	$\Delta\%$	RMSE	$\Delta\%$
NORMAL (full model)	0.007225	+3.69	0.004925	+4.37	0.005035	+3.45
GATE_SHUFFLE	0.007106	+1.98	0.004844	+2.65	0.005090	+4.58
EXPERT_SHUFFLE	0.007044	+1.09	0.004785	+1.39	0.004925	+1.20
BOTH_SHUFFLE	0.007144	+2.53	0.004765	+0.97	0.004891	+0.50
SINGLE (pooled ridge)	0.006968	+0.00	0.004719	+0.00	0.004867	+0.00

Table 4: Protocol ablation. Improvement of the regime-conditioned model over its *matched* pooled-ridge baseline, as four evaluation shortcuts are removed cumulatively. Positive means the regime model looks better. p is a HAC-corrected Diebold–Mariano test against that baseline. At $h = 1$ the aliased and realized-volatility targets coincide by construction, so P3 and P4 are identical.

Correction applied		$h = 1$		$h = 5$		$h = 20$	
		$\Delta\%$	p	$\Delta\%$	p	$\Delta\%$	p
P0	our original implementation	+1.93	<0.001	+1.25	0.001	+0.63	0.183
P1	+ tuned/standardized baseline	-1.03	0.041	-0.37	0.123	-0.23	0.368
P2	+ train/test embargo	-1.06	0.036	-0.34	0.152	-0.19	0.447
P3	+ nested selection	-3.69	0.001	-1.02	0.004	-0.76	0.014
P4	+ realized-vol target	-3.69	0.001	-4.37	0.028	-3.45	<0.001

small but detectable benefit. SPY lands among the cases where it does not.

Both readings agree on the underlying picture. The regime label restates trailing realized volatility at coarse resolution, and the feature set already carries that as $RV^{(1)}$, $RV^{(5)}$ and $RV^{(22)}$. A three-level discretization of a variable the model reads continuously adds little, and fragmenting the training data across experts to exploit it costs more than it adds. Section 6.4 puts numbers on both sides of that trade.

6.3 Protocol ablation

Table 4 and Figure 1 trace the improvement of the regime model over its matched pooled-ridge baseline as four methodological choices are corrected in sequence. The model is unchanged throughout; only the protocol moves.

P0, the starting point, is our own first implementation, written before we applied any correction in this paper. It reconstructs no published study, and we make no claim that any particular paper combines all four choices. Our claim is narrower and more useful: each of the four is easy to make, each survived our own code review, and together they produce a significant positive result for a method that does not work.

P0 uses the aliased target, an unstandardized fixed-penalty baseline, no embargo, and K and the gating rule chosen by out-of-sample

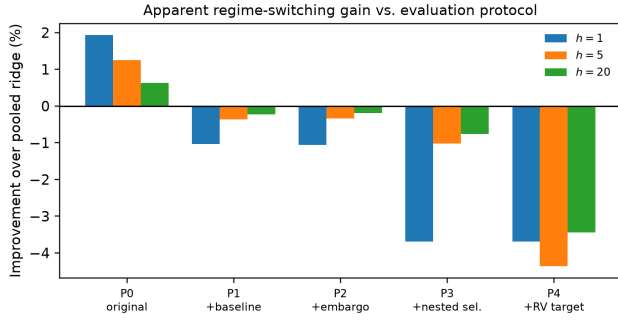


Figure 1: The apparent gain from regime conditioning as a function of the evaluation protocol, not of the model. The model is identical in every bar; only the protocol changes.

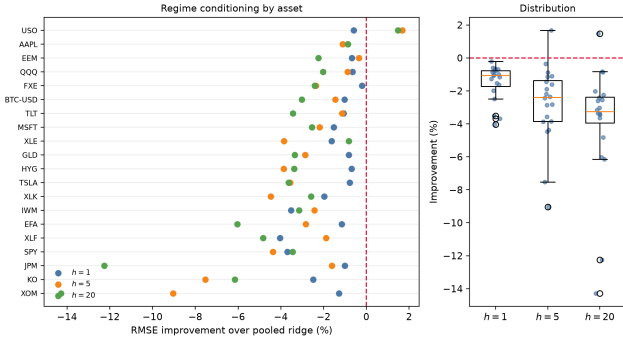


Figure 2: Cross-asset replication. Each point is one asset at one horizon; positive would favour regime conditioning. 58 of 60 cells are negative.

score. Under it the regime model improves on its baseline by 1.9% at $h = 1$ and 1.3% at $h = 5$, both significant at the 1% level. Correcting the baseline alone (P1) flips the sign at every horizon. Adding the embargo (P2) changes little here. At these horizons the leak is real but small, and we keep it because nobody can know its size in advance, not because it dominates. Nested selection (P3) removes a further 0.6 to 2.6 percentage points, the largest single correction. The corrected target (P4) leaves the regime model between 3.4% and 4.4% behind the pooled ridge, significantly so at every horizon.

They compound for a reason. All four widen the flexibility of the proposed model relative to its benchmark, so their effects share a sign, which is also why none of them looks alarming alone. The literature documents each failure mode separately: leakage across a train/test boundary [18, 21], selection reported as evaluation [3, 15], and under-specified baselines [9]. The interaction turns a -4% result into a significant $+2\%$ one, and a checklist applied one item at a time will miss interactions.

6.4 Cross-asset replication

A single-asset null carries weak evidence, and SPY is the least favourable series we could have picked. It is the most liquid and most-studied series available, and the one where trailing realized

Table 5: Cross-asset replication. Each cell is the out-of-sample RMSE improvement of the regime-conditioned model over the pooled ridge fitted on identical features, in percent; positive would favour regime conditioning. $\dagger$ marks a Diebold–Mariano rejection at the 5% level, always in the direction of the pooled ridge. Assets are sorted by mean trailing volatility.

Ticker	Class	Vol	$h=1$	$h=5$	$h=20$
HYG	High yield credit	0.0048	-0.69	-3.87	-3.38
FXE	EUR/USD	0.0051	-0.20	-2.35 $\dagger$	-2.42
TLT	Long treasuries	0.0083	-1.07 $\dagger$	-1.14	-3.44
SPY	US large cap	0.0096	-3.69 $\dagger$	-4.37 $\dagger$	-3.45 $\dagger$
KO	Single name (staples)	0.0098	-2.48 $\dagger$	-7.53 $\dagger$	-6.15
GLD	Gold	0.0102	-0.81 $\dagger$	-2.86 $\dagger$	-3.35 $\dagger$
EFA	Developed ex-US	0.0108	-1.16	-2.83 $\dagger$	-6.04 $\dagger$
QQQ	US tech-heavy	0.0117	-0.66 $\dagger$	-0.88	-2.03
XLK	Technology	0.0120	-1.97 $\dagger$	-4.47 $\dagger$	-2.60 $\dagger$
IWM	US small cap	0.0130	-3.52 $\dagger$	-2.43 $\dagger$	-3.15 $\dagger$
XLF	Financials	0.0136	-4.04 $\dagger$	-1.88	-4.83
EEM	Emerging markets	0.0141	-0.67	-0.35	-2.25 $\dagger$
XOM	Single name (energy)	0.0142	-1.28 $\dagger$	-9.04 $\dagger$	-14.28 $\dagger$
MSFT	Single name (tech)	0.0149	-1.53 $\dagger$	-2.19 $\dagger$	-2.54 $\dagger$
XLE	Energy	0.0157	-1.63 $\dagger$	-3.85 $\dagger$	-0.82
JPM	Single name (financial)	0.0172	-1.00	-1.61	-12.26 $\dagger$
AAPL	Single name (tech)	0.0177	-0.86 $\dagger$	-1.10 $\dagger$	-0.87
USO	Crude oil	0.0201	-0.60	+1.67	+1.48
BTC-USD	Bitcoin	0.0302	-1.03	-1.45 $\dagger$	-3.03 $\dagger$
TSLA	Single name (high vol)	0.0324	-0.78 $\dagger$	-3.57 $\dagger$	-3.64 $\dagger$
Median			-1.05	-2.39	-3.25

volatility tells you most, leaving regime labels the least room to add anything. We therefore repeat the corrected protocol on 20 assets spanning broad equity indices, sectors, single names, long-duration rates, high-yield credit, commodities, FX and crypto, with sample histories of 12 to 21 years.

The null holds across the cross-section. Table 5 and Figure 2 report the regime model’s improvement over its matched pooled ridge: 58 of 60 asset-horizon cells come out negative, with medians of -1.05% , -2.39% and -3.25% at $h = 1, 5, 20$. Of the 60 cells, 37 are significant deteriorations under a Diebold–Mariano test and *none* is a significant improvement. Both positive cells belong to crude oil (USO), and neither reaches significance. Correlations between an asset’s mean volatility and the size of the effect stay small and flip sign across horizons ($+0.14$, $+0.20$, $+0.05$), so we find no sign that regime conditioning pays for more volatile or less well-behaved assets.

6.5 The value of regime timing

The cross-section measures what the single-asset ablation could only bound (Figure 3). Pooling all 60 pairs, correct regime timing beats shuffled timing in 37 of them. A sign test misses significance at $p = 0.09$, while the paired magnitudes reach it: the mean benefit is $+0.69$ percentage points of RMSE, and a Wilcoxon signed-rank test on the paired differences gives $p = 0.023$. The benefit grows with horizon, from $+0.23$ points at $h=1$ to $+0.54$ at $h=5$ and $+1.29$ at $h=20$, the pattern we would expect if the regime state carries information about persistent volatility levels rather than about tomorrow.

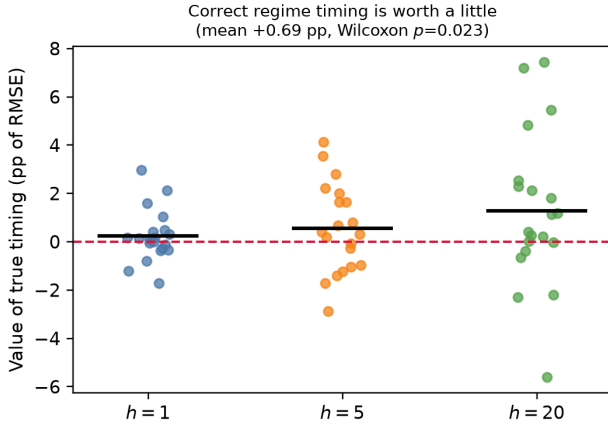


Figure 3: The paired benefit of correct regime timing over a time-shuffled gate, in percentage points of RMSE, one point per asset-horizon pair; bars are means. The benefit is small, positive on average, and grows with horizon.

Table 6: Every way of using the regime signal, against the pooled ridge, over 60 asset-horizon pairs. “Median” is the median improvement in percent (positive favours the row). “Wins” counts pairs where the row beats the pooled ridge; “sig. better” and “sig. worse” count Diebold–Mariano rejections at 5%. No regime variant achieves a single significant improvement in 240 tests, whereas GARCH and EWMA—which ignore regimes entirely—beat the pooled ridge more often than not.

Model	Median	Wins	sig. better	sig. worse
Regime gating (full mixture)	-2.30%	2/60	0	37
with time-shuffled gate	-3.14%	3/60	0	40
Regime gating, shrunk to pooled	-0.72%	11/60	0	18
Regime posteriors as features	-0.63%	4/60	0	31
HAR	-0.13%	26/60	7	8
GARCH(1,1)	+0.59%	39/60	4	3
EWMA	+0.42%	33/60	14	9

Regime timing therefore carries some value. Two questions remain: whether it carries enough, and whether the gating mechanism blocks it.

6.6 Using the regime signal without gating

If the partition binds, then using the regime posteriors *without* partitioning should recover the loss. We test the two obvious ways to do that, both under the same protocol. The first appends the filtered posteriors $\gamma_t(1), \dots, \gamma_t(K)$ to the feature matrix of a single pooled ridge, handing the model the regime signal at zero partitioning cost. The second shrinks the gated mixture toward the pooled fit by a factor selected on the inner validation block, over a grid containing both endpoints, so it can land on either extreme.

Neither route rescues the method. Table 6 reports all four. Feature augmentation loses by 0.63% at the median and shrinkage by

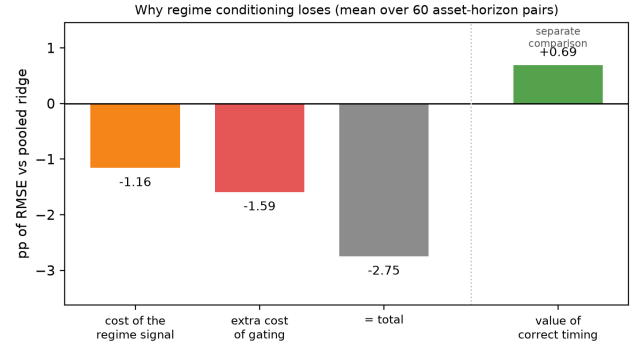


Figure 4: Decomposition of the regime model’s loss against the pooled ridge. The first two bars are additive by construction. The timing term measures a different comparison, gated versus gated-with-shuffled-gate, and forms no part of the total.

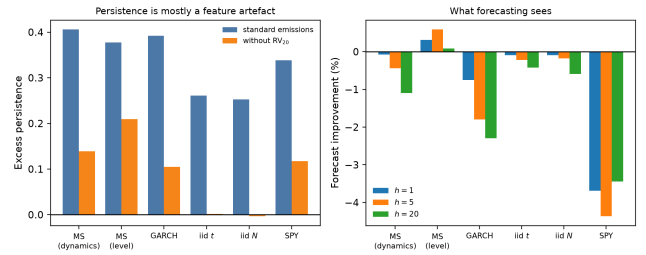


Figure 5: Left: apparent regime persistence, with the standard emission vector and with the 20-day rolling volatility removed. On i.i.d. data the first is large and the second is zero. Right: forecast improvement from conditioning on the inferred state. Only the right panel separates the worlds that contain regimes from those that do not.

0.72%, against 2.30% for full gating. Across the four variants and 240 Diebold–Mariano tests, *none* improves on the pooled ridge at any conventional level.

Figure 4 splits the two costs. Injecting the regime signal at all costs 1.16 points of RMSE on average (Wilcoxon $p = 2.4 \times 10^{-9}$); gating on it rather than conditioning on it costs a further 1.59 (paired Wilcoxon $p = 8.7 \times 10^{-8}$). Both costs are real, and the partition is the larger, so the mechanism does matter. Remove it and the regime model still trails a plain ridge.

Two details deserve recording. The shrinkage grid includes $\lambda = 1$, the pooled ridge itself, and the inner validation block picks it in 15% of folds, settling on a mean shrinkage of 0.45 instead. In sample the regime signal looks useful often enough to get selected; out of sample it does not deliver. Second, the benchmarks that ignore regimes outperform the ridge we use as the reference: GARCH(1,1) beats it in 39 of 60 pairs and EWMA in 33. The regime model loses to a control that is not even the strongest baseline available.

Table 7: What the HMM learns from each world. All rows use the identical pipeline. “Separation” is the ratio of the highest to the lowest state’s mean realized volatility; “excess persistence” is the average self-transition probability minus the stationary probability, so zero means states are temporally independent. Persistence is reported with the standard emission vector and again with the 20-day rolling volatility removed; on data with no dynamics at all the first is large and the second is zero, so most of what the usual diagnostic reports is autocorrelation contributed by an overlapping-window feature. “State recovery” is the adjusted Rand index against the true latent path where one exists. The final columns are the out-of-sample RMSE improvement from conditioning on the inferred state. Separation and persistence do not distinguish the worlds containing regimes from those that do not; forecasting does.

Data-generating process	True regimes?	Separation	Excess persistence		State recovery	Forecast improvement		
			std.	no RV ₂₀		h=1	h=5	h=20
Markov switching (differing dynamics)	✓	2.80	+0.406	+0.139	0.13	-0.07%	-0.44%	-1.10%
Markov switching (differing level)	✓	3.15	+0.378	+0.209	0.33	+0.31%	+0.59%	+0.09%
GARCH(1,1)-t	×	3.00	+0.392	+0.105	–	-0.75%	-1.79%	-2.29%
i.i.d. Student-t	×	1.60	+0.261	+0.002	–	-0.09%	-0.22%	-0.42%
i.i.d. Gaussian	×	1.28	+0.253	-0.003	–	-0.09%	-0.17%	-0.59%
SPY (real data)	–	2.51	+0.339	+0.118	–	-3.69%	-4.37%	-3.45%

6.7 What the HMM learns from returns

A null result on real data admits two readings on its own: markets may have no regimes, or our pipeline may fail to find regimes that are there. We settle it by running the identical pipeline on synthetic data from worlds whose answer we know. Two have genuine Markov switching, one has persistent continuous volatility and no discrete states, and two have no dynamics at all. We calibrate the processes to SPY’s unconditional volatility, excess kurtosis and absolute-return autocorrelation, and we give the GARCH world Student- t innovations so that it stands in for real returns rather than separating on tail thickness alone.

Table 7 carries four findings.

The usual descriptive diagnostics do not identify regimes. State separation reaches 3.00 in the GARCH world, which contains no discrete states, against 3.15 for genuine two-state switching. Show a reader the fitted means, the transition matrix and a shading plot, and they cannot tell the two apart. Both resemble the real data.

The emission vector manufactures most of the reported persistence. Fit the HMM to i.i.d. Gaussian returns, with no volatility clustering and no regimes, and it reports excess persistence of +0.25, a self-transition probability double the stationary probability. Remove the 20-day rolling volatility from the emission vector and the figure falls to -0.003, the zero it should have shown all along. Consecutive values of $RV^{(20)}$ share 19 of their 20 observations, so the filter reads back autocorrelation that our own feature construction supplied. The size of the artefact matches the size of the real signal: of SPY’s apparent +0.339, only +0.118 survives the removal, and of the GARCH world’s +0.392, only +0.105. Since “the inferred regimes are highly persistent” ranks among the most reported diagnostics in this literature, we encourage reporting it in the ablated form.

Only forecasting discriminates, and it places real data with GARCH. Regime conditioning helps in the world where regimes exist and differ in level (+0.31% and +0.59% at $h = 1, 5$, both significant across 12 replications), and hurts in the three worlds where they do not. Real SPY sits with GARCH rather than with Markov switching. Our

pipeline detects regimes when they are present, so the positive control works, which licenses reading the empirical null as a statement about markets and not about our implementation.

An unexpected asymmetry. We built the “differing dynamics” world as the *strongest* case for a regime-conditioned mixture. Its two states share a volatility level and differ in how fast volatility mean-reverts, so per-regime experts have different coefficients to learn. Regime conditioning fails there (-0.44% at $h = 5$), and the state-recovery column explains it: the adjusted Rand index against the true path is 0.13, against 0.33 in the world where states differ in level.

The estimator causes this, not our setup. Baum–Welch maximises the likelihood of the *emissions*, so it identifies states that differ in their emission distribution and stays close to blind to states that differ in how the process evolves. A state that differs in level is one that trailing realized volatility already reveals, while a state that differs in dynamics is the one that would carry new information. The estimator excels at finding the regimes least worth conditioning on and struggles with the ones that would help. Its objective and the forecasting objective are different, and a correct implementation cannot close that gap.

6.8 Two fixes the diagnosis implies

The two failures above, an emission vector that manufactures persistence and an objective that finds the wrong states, each suggest a remedy. We test both.

Emissions without overlapping windows. Replacing the emission vector with volatility-standardised returns $z_t = r_t / \hat{\sigma}_{t-1}$, where $\hat{\sigma}$ is a causal EWMA of past squared returns, removes the manufactured persistence. Excess persistence on i.i.d. Gaussian returns falls from +0.301 to -0.005, and on i.i.d. Student- t from +0.297 to +0.002. The diagnostic then starts to *discriminate*: the GARCH world, holding no discrete states, falls to +0.013, while the two genuine regime worlds hold +0.065 and +0.048. Under any process whose only structure is time-varying scale, the standardised series approaches

Table 8: Two-stage versus jointly-estimated regimes. Both are RMSE improvement over the same pooled ridge, in percent, averaged over four replications for the synthetic worlds. Both arms use the identical protocol: embargoed walk-forward splits, K and the ridge penalty selected on the inner validation block, and causal prediction. Joint estimation is much better where regimes exist and differ in level, no better where they differ only in dynamics, and markedly worse where there are no regimes to find.

Data-generating process	Two-stage			Joint		
	$h=1$	$h=5$	$h=20$	$h=1$	$h=5$	$h=20$
Markov switching (level)	+0.37	+0.72	-0.66	+2.62	+3.91	-6.97
Markov switching (dynamics)	+0.13	+0.14	-0.70	-1.66	-0.98	-4.54
GARCH(1,1)- t (no regimes)	-0.98	-3.52	-6.44	-4.80	-7.38	-11.84
SPY (real data)	-1.13	-4.58	-3.56	-8.10	-11.41	-2.72

i.i.d., so persistence that survives points to structure the volatility level leaves unexplained.

Applied to SPY, the diagnostic returns -0.001 . Divide out their own volatility level and real equity returns show *less* regime persistence than a GARCH process containing no regimes. The apparent regime structure in Figure 7 is the volatility level and nothing besides. This is the cleanest statement the paper can make, and we measured it rather than inferring it.

Estimating the states jointly with the forecast. If Baum–Welch optimises the wrong objective, the fix is to make the emission the predictive density itself, $p(y_t | x_t, z_t=k) = \mathcal{N}(\beta'_k x_t, \sigma_k^2)$, and estimate β , σ and the transition matrix jointly by EM. Hamilton [12] introduced this Markov-switching regression, so the estimator is not new. We contribute the controlled comparison against the two-stage pipeline, on data whose regimes we control, giving both arms the same protocol: embargoed splits, K and the ridge penalty selected on the same inner validation block, and a horizon-aware causal filter for out-of-sample prediction.

The results split three ways.

Joint estimation gains a great deal where regimes exist and the estimator can identify them. On the level-switching world it returns +2.62% and +3.91% at $h = 1, 5$ against the two-stage pipeline’s +0.37% and +0.72%, a sevenfold difference. Our diagnosis holds up that far: the objective mismatch is real, and removing it recovers much more of the available signal.

Joint estimation fails to rescue the dynamics-switching world, the case we expected it to fix. It loses there at every horizon and beats the two-stage pipeline in 1 of 12 replications. We predicted the opposite. The implication runs further than what we set out to demonstrate: states that share a volatility level and differ in how volatility evolves appear unrecoverable from a return series by *either* estimator. That bounds what any regime detector reading returns alone can deliver, and it constrains the data rather than the choice of objective.

In the GARCH world and in real data, holding no regimes, joint estimation trails the two-stage pipeline by a wide margin (-4.80% and -8.10% at $h = 1$). Give a model K times the parameters and nothing to find, and the extra flexibility overfits. This matches the

Table 9: Model Confidence Set [14] at the 90% level, computed separately for each of the 60 asset-horizon pairs with a stationary block bootstrap at block length h . Entries are the percentage of pairs in which the model survives into the set of models that cannot be distinguished from the best. Regime-conditioned models are in bold. Unlike the pairwise tests, this accounts for the full set of competitors simultaneously.

Model	MSE	QLIKE
Ridge (pooled)	75%	67%
HAR	52%	65%
GARCH(1,1)	78%	88%
EWMA	77%	85%
Gradient boosting (pooled)	18%	43%
Regime gating	28%	42%
Regime as features	40%	55%
Regime gating, shrunk	60%	72%
Regime gating, GBM experts	15%	35%

emission diagnostic above: SPY holds no residual regime structure for a better estimator to exploit, so the more powerful estimator does more damage.

One practical recommendation follows. Run the standardised-emission diagnostic first. If it returns near zero, as it does for SPY, no regime machinery of any sophistication will help; condition on volatility continuously and stop. If it returns something well above zero, prefer joint estimation to the two-stage pipeline, and expect it to find only regimes that differ in level.

6.9 Model confidence sets

Our tests so far run pairwise. Against a dozen competitors that asks too little: a model can avoid losing to any single rival and still drop out once all of them compete at once. The Model Confidence Set [14] returns the set of models that cannot be distinguished from the best, the object our claim concerns. We compute it separately for each of the 60 asset-horizon pairs at the 90% level, using a stationary block bootstrap [26] with block length h to preserve the dependence that overlapping targets induce.

Table 9 reports how often each model survives, and the ordering leaves no ambiguity. GARCH(1,1) and EWMA, neither of which knows anything about regimes, survive in 78% and 77% of pairs under MSE and in 88% and 85% under QLIKE. The full regime mixture survives in 28% and 42%. Boosted regime experts, the most elaborate model we fit, survive least often of all, at 15% and 35%.

The shrunk mixture holds up better than the other regime variants, at 60% and 72%, and the reason deflates the result: shrinkage selected on the inner split pulls it most of the way back toward the pooled ridge, so the ridge is what survives. Across all four regime variants and both losses, no variant survives as often as the simplest volatility model in the comparison.

6.10 Equivalence bounds

An underpowered test also fails to reject equal accuracy, so that failure on its own establishes nothing. Since our central claim is a

Table 10: Equivalence bounds. For each asset-horizon pair we form a 95% confidence interval on the regime model’s RMSE improvement over the pooled ridge and report the upper limit, i.e. the largest improvement the data cannot exclude. A negative bound means *no* improvement of any size is consistent with the data at 95%. This is the statement a null result requires: failing to reject equal accuracy is also what an underpowered test produces.

Variant	Horizon	Median bound	Worst case	Bound < 0
Regime gating (full mixture)	$h = 1$	-0.12%	+0.80%	13/20
	$h = 5$	-0.47%	+6.14%	13/20
	$h = 20$	-0.11%	+10.34%	11/20
Regime posteriors as features	$h = 1$	-0.02%	+0.64%	11/20
	$h = 5$	-0.15%	+4.39%	11/20
	$h = 20$	+0.22%	+3.89%	9/20
Regime gating, shrunk to pooled	$h = 1$	+0.13%	+0.94%	6/20
	$h = 5$	+0.46%	+6.52%	7/20
	$h = 20$	+0.50%	+6.94%	5/20

Table 11: Sensitivity to the HMM’s random initialization. Ten seeds, everything else fixed. $\Delta\%$ is improvement over the (deterministic) pooled ridge. The spread across seeds is comparable to the effect size the comparison is trying to resolve.

Horizon	Pooled ridge	RMSE mean	RMSE s.d.	$\Delta\%$ min	$\Delta\%$ max
$h = 1$	0.006968	0.007114	0.000081	-3.69	-0.18
$h = 5$	0.004719	0.004948	0.000059	-6.71	-1.94
$h = 20$	0.004867	0.005043	0.000046	-5.19	-2.36

null, we carry the burden of showing the study could have detected an effect had one existed.

Table 10 reports, for each asset-horizon pair, the upper limit of a 95% confidence interval on the regime model’s RMSE improvement over the pooled ridge: the largest improvement the data leave open. We build the interval on the same HAC-corrected loss differential as the Diebold–Mariano tests and map it monotonically onto the RMSE percentage scale.

In the median pair that upper limit comes out negative at every horizon, so no improvement of any size survives at 95%. The bounds tighten at $h=1$, where even the least informative asset rules out improvements above 0.80%. They loosen at longer horizons, and the weakest single case at $h=20$ rules out only improvements above 10.3%, because overlapping 20-day targets leave far fewer effective observations and the HAC correction reflects that. We claim no tight bound at $h=20$ for every asset. We claim that at $h=1$, and in the median case at all horizons, we measured the null instead of failing to find one.

6.11 Seed sensitivity

Baum–Welch maximizes a non-convex likelihood. Table 11 and Figure 6 report the spread of out-of-sample RMSE across ten random initializations, holding everything else fixed. That range matches the differences the model comparison sets out to resolve, and the seed also shifts the number of regimes that nested selection picks. A single-seed comparison between a regime model and a baseline

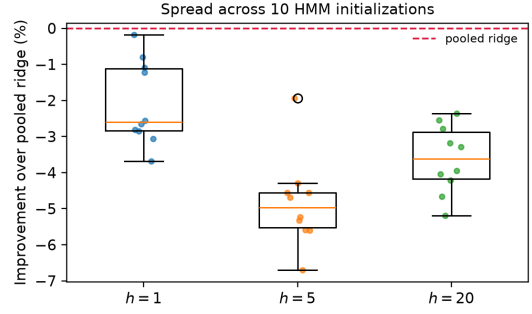


Figure 6: Improvement over the pooled ridge across ten HMM initializations. Every seed is negative, but the spread is wide enough that a single run could report anything within it.

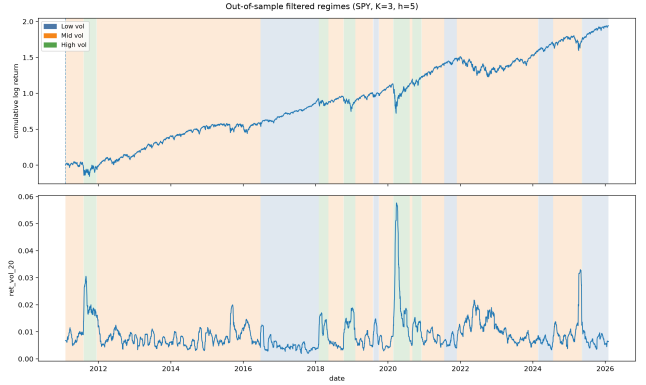


Figure 7: Out-of-sample filtered regimes for the $K=3$ model at $h=5$. Shading marks contiguous hard-state segments with opacity proportional to posterior confidence. Top: cumulative log return. Bottom: trailing realized volatility. The high-volatility state aligns with 2011, 2015–16, 2018, 2020 and 2022.

therefore tells you little, and single-seed results remain the norm in this literature.

Every seed leaves the regime model behind the pooled ridge, so our conclusion survives the draw. The spread shows that the *magnitude* any one run reports belongs to that run and not to the method, and that a study reporting a small positive effect from a single seed has measured nothing. Report the seed distribution as a matter of course.

6.12 The regimes are real

The HMM does find structure. Figure 7 shows the filtered high-volatility state tracking recognized stress episodes, and doing so causally, without lookahead. The fits behave well: across all 15 folds no state collapses, the canonical volatility ordering stays monotone in every fold for $K \in \{2, 3\}$, and mean self-transition probabilities reach 0.93, 0.81 and 0.68 for $K = 2, 3, 4$, so the inferred states persist instead of flickering.

Table 12: What the inferred regimes do capture ($K=3$, $h=5$, out-of-sample). “Stress” is the top decile of realized volatility over the forecast window, so the unconditional rate is 10%. Lift is the ratio of the within-regime rate to that base rate. The high-volatility state occupies a sixth of the sample and contains nearly three-fifths of all stress events—the regimes are genuinely informative, just not incrementally so for the conditional mean.

Regime	% of time	Mean RV ⁽²⁰⁾	P(stress)	Lift	% of all stress
Low vol	47.3	0.00623	0.021	0.21	9.8
Mid vol	36.4	0.00884	0.087	0.87	31.5
High vol	16.3	0.01707	0.360	3.60	58.7

Table 12 quantifies it, and the numbers are large. The high-volatility state occupies 16% of the out-of-sample period and contains 59% of all volatility spikes. Conditional on sitting in it, the probability of a top-decile spike over the next week is 36% against an unconditional 10%, a lift of 3.6 $\times$. The causal HMM works as a stress classifier.

This finding takes the paper’s result to its sharpest form, and it admits a direct test. If the regime label re-encodes trailing volatility, then trailing volatility should recover it. Regressing the filtered hard state on the three HAR components alone, with no other features and no HMM, recovers it with 95.7% accuracy against a 46.2% base rate. (We fit this in sample as a description, not a forecast. The question is whether the regime variable is a function of information already present, not whether anyone can predict it.)

That mechanism drives Section 6.6. The HMM state encodes, at three noisy levels refitted once a year, a quantity every model in the comparison already reads continuously and without estimation error. It cannot supply information the model lacks, and it does supply estimation noise, which is why feature augmentation loses even with no partition to pay for. Gating then adds a second and larger cost.

This distinction matters in practice. Regime labels may still earn their place in communication, risk limits or position sizing, and a 3.6 $\times$ lift on stress is a usable signal for those purposes. Our result covers one thing: they do not improve the conditional mean of realized volatility.

7 A Checklist

The audit that produced this paper reduces to a small number of checks that we would encourage as standard for regime-switching forecasting work:

- (1) **Verify the target varies with the horizon.** Compare the marginal distributions of $y^{(h)}$ across h . If they are identical, the target is a single-period quantity at longer lead, not a multi-horizon forecast.
- (2) **Use filtered, not smoothed, state probabilities,** and verify it by checking that y_t holds unchanged when you delete data after t deleted.
- (3) **Purge the training window by the horizon.**
- (4) **Select hyperparameters inside the training window.** If the number of regimes is chosen on the test score, the

test score is no longer an estimate of out-of-sample performance.

- (5) **Give the baseline the same care as the model:** standardization, tuned regularization, and access to the same features. Include HAR and GARCH for volatility targets.
- (6) **Run a gate-shuffle ablation,** and compare what correct timing buys against what the conditioning mechanism costs. Both quantities are measurable on the same scale, and only their difference is the result. Interpret the ablation across many series: on a single asset it is too noisy to distinguish “no signal” from “signal too small to pay for itself”.
- (7) **Check that an ablation is not a duplicate.** Ours were: two of the four arms in our first design returned bitwise-identical numbers.
- (8) **Relabel states canonically before pooling across folds,** or every per-regime table mixes different regimes together without warning.
- (9) **Check the regime variable is not redundant** with a feature the model already has. Regress the inferred state on your existing volatility features: ours was 95.7% recoverable, and nothing downstream could fix that.
- (10) **Report persistence with overlapping-window emissions removed.** An HMM fitted to i.i.d. noise reports substantial regime persistence if its emission vector contains a rolling statistic. Refit without it and report both numbers.
- (11) **Validate on a simulated world with known regimes.** If conditioning does not help where regimes provably exist, the architecture is at fault and no amount of real data will reveal it. If it does help there and not on real data, that is evidence about markets rather than about code.
- (12) **Test the signal without the mechanism.** Before concluding that gating is what fails, feed the posteriors in as plain features. The two diagnoses imply different remedies.
- (13) **Report the seed distribution,** not a single run.
- (14) **Test significance** with a HAC-corrected Diebold–Mariano test and correct for the number of comparisons.
- (15) **Report what your null excludes,** not just that you failed to reject. An equivalence bound distinguishes a measured null from an underpowered one.

8 Limitations

Our result holds along specific dimensions, and we would rather not overgeneralize from them.

Asset universe. We cover 20 liquid, exchange-traded instruments. They span asset classes, but all of them trade liquidly and carry long histories. Illiquid or short-history assets, and instruments whose volatility follows scheduled events instead of persistence, fall outside our sample.

Expert families. We test ridge and gradient-boosted experts. The gating cost depends on the family: large and significant for ridge, indistinguishable from zero for boosting (Section A.1). The decomposition in Section 6.6 therefore describes linear experts, and we would not assume it generalises. Across both families the regime-conditioned model loses to a plain pooled ridge.

One family of regime detector. We use a Gaussian HMM on a three-dimensional emission vector, every component of which is

a function of past returns. A detector reading information outside the trailing-volatility span, such as order flow, options-implied measures or macroeconomic releases, would escape the redundancy that Section 6.6 documents in ours. We regard that as the most promising direction the paper points at, and the redundancy diagnostic of Section 6.6 as the first check such work should report.

Economic evaluation is narrow. We test volatility targeting and VaR one asset at a time. Regime information may matter more for problems our setup cannot see: cross-asset allocation, where correlations shift with regime [11], or drawdown control, where the timing of a regime call outweighs the accuracy of a conditional mean.

Gaussian HMM with a small emission vector. Richer emissions, non-Gaussian observation densities or duration-dependent transitions could recover regimes less redundant with trailing volatility.

Conditional mean only. We evaluate point forecasts. Regime models produce predictive *distributions*, and their value may sit in tail quantiles or density forecasts, which our loss functions leave unprobed.

Fixed evaluation window. Our results cover 2011–2026 out of sample. That period contains major stress episodes and remains one realization of history.

Pairwise testing. Section 6.9 reports a Model Confidence Set, but our headline comparisons use Holm-corrected pairwise Diebold–Mariano tests against a fixed reference model, since our claims name specific benchmarks. The two agree here, and readers who prefer the joint object should read Table 9 as the primary evidence.

9 Conclusion

We set out to show that causally-inferred market regimes improve volatility forecasting, and found that they do not. Against well-specified benchmarks, with an embargoed split and hyperparameters selected inside the training window, a pooled ridge beats the HMM regime-conditioned version of itself in 58 of 60 asset-horizon pairs, and loses significantly in none.

The reason it fails matters more, and it differs from the one we first reached for. The regime signal exists: the inferred high-volatility state predicts stress with a $3.6\times$ lift, and correct timing earns a measurable 0.69 points of RMSE. We first read the gating mechanism as too expensive, which predicts that feeding the posteriors in as plain features should work. That variant loses too, by 1.16 points before any gating cost arrives. Features the model already holds recover the regime state with 95.7% accuracy, so every route of injecting it supplies estimation noise and no information, and gating adds a further 1.59 points.

The distinction changes what anyone should do next. “The mechanism is too expensive” invites better mechanisms. “The variable duplicates what you already condition on” invites a different regime detector, one reading information outside the trailing-volatility span, or none at all. Without testing the non-gating variants we could not have separated the two, and we would have published the wrong recommendation.

The conclusion held on every axis we pushed it: nonlinear experts, a Model Confidence Set weighing all competitors at once, and a decision-relevant loss. That last case softens the picture, and we report it because it is true and because it shows that the ranking

under squared error need not be the ranking that matters. Under volatility targeting the regime models become indistinguishable from the pooled ridge instead of trailing it. The recommendation stands, since the models ignoring regimes stay significantly ahead of both.

We also recovered a significant positive result by reintroducing four common evaluation shortcuts. Negative results in forecasting resist interpretation on their own, because anyone can attribute a null to insufficient tuning. Tracing the path from a null to a significant positive and back makes the failure mode concrete, and we hope harder to reproduce by accident.

Reproducibility

Three entry points produce every result from raw data: the study driver, the protocol ablation and the seed sweep. Every table and figure in this paper comes from the stored artifacts programmatically, with nothing transcribed by hand. This buys more than tidiness. The project this paper grew out of carried a hand-copied results table whose numbers matched no artifact in the repository, and automating the path from artifact to table is what caught it.

Tests cover the correctness claims that matter, including the lookahead check described in Section 5: we recompute the filtered probabilities on truncated inputs and assert equality, so a regression to smoothed posteriors fails the suite instead of improving the results without anyone noticing. Code, configuration and artifacts accompany the submission.

A Additional robustness checks

Both checks below confirm the conclusions of Section 6 without altering them, and are reported here to keep the main text on the argument rather than on its corroboration.

A.1 Nonlinear experts

Our experts are linear, which invites the objection that a regime label earns its keep only when it gates different functional forms. We therefore repeat the comparison with gradient-boosted experts, pooled and gated, holding K fixed at 3.

The objection fails, though for a reason we did not expect. Gradient boosting scores *worse* than ridge at this task, by a median 6.03% in pooled RMSE, and it wins 0 of 60 pairs. That matches the long-standing observation that realized volatility runs close to linear in its own lagged components, the fact HAR builds on [6]. Gating those experts leaves them at 6.29%, so the branch stays uncompetitive whatever we do with the regime signal.

The gating cost carries the interest here. With ridge experts, gating costs 2.30 points (Wilcoxon $p = 1.6 \times 10^{-10}$); with boosted experts it costs 0.11 points, indistinguishable from zero ($p = 0.37$). An earlier draft of ours predicted the reverse, reasoning that more flexible experts would raise the partition cost since each one fits on a fraction of the window. That prediction was wrong. A boosted model can represent regime-dependent behaviour internally through its splits, which makes explicit gating redundant instead of costly, and its own variance swamps any marginal partition effect. We record the failed prediction because it bears on the paper’s method: the partition cost belongs to the expert family and does not generalise, so our decomposition in Section 6.6 describes linear experts.

Table 13: Economic evaluation, median over 20 assets. Volatility targeting sizes a position at $\sigma^{\text{target}}/\hat{\sigma}_t$ for a 10% annualised target; the reported error is $|\text{realised} - \text{target}|$ annualised volatility, so lower is better. VaR is at the 5% level with the forecast-to-quantile scaling calibrated by filtered historical simulation on a trailing window, evaluated at $h=1$ where the realised signed return is unambiguous. “Pass CC” is the share of assets passing Christoffersen conditional coverage at 5%.

Model	Vol-target error			VaR (5%, $h=1$)	
	$h=1$	$h=5$	$h=20$	Rate	Pass CC
Ridge (pooled)	0.0375	0.0104	0.0049	0.052	95%
HAR	0.0347	0.0099	0.0040	0.052	95%
GARCH(1,1)	0.0337	0.0093	0.0031	0.052	90%
EWMA	0.0350	0.0109	0.0040	0.052	75%
Gradient boosting (pooled)	0.0389	0.0116	0.0074	0.053	65%
Regime gating	0.0370	0.0111	0.0056	0.053	95%
Regime as features	0.0369	0.0119	0.0056	0.053	95%
Regime gating, shrunk	0.0360	0.0099	0.0046	0.051	95%
Regime gating, GBM experts	0.0392	0.0130	0.0085	0.053	65%

A.2 Economic evaluation

Nobody uses a volatility forecast to minimise squared error. A model with worse RMSE but a $3.6\times$ lift on stress classification could still feed a risk system better, and that would give this paper its one positive finding. We test two standard uses.

Volatility targeting. We size the position at $\sigma^{\text{target}}/\hat{\sigma}_t$ for a 10% annualised target [23] and measure how far realized volatility lands from that target. The exercise assumes no distribution and the loss reads off the result.

Value-at-Risk. We convert the forecast to a 5% quantile by filtered historical simulation on a trailing window, so the scaling uses past information only, then test coverage with Kupiec [20] and Christoffersen [5].

The economic evidence in Table 13 runs softer than the statistical evidence, and in one respect kinder to regime conditioning. On volatility targeting no regime variant differs significantly from the pooled ridge. The full mixture comes out marginally worse (paired Wilcoxon $p = 0.053$) and the shrunk mixture marginally better ($p = 0.34$), neither conclusive. Under a decision-relevant loss the regime models become *indistinguishable* from the pooled ridge instead of trailing it, as they do under RMSE. That difference between loss functions deserves recording.

It still builds no case for regime conditioning, because the regime-free benchmarks beat both. GARCH(1,1) cuts the volatility-targeting error relative to the pooled ridge at every horizon ($p < 10^{-4}$, better in 43 of 60 pairs), and HAR does the same ($p = 0.006$). The regime models tie with the pooled ridge, and the pooled ridge is not the frontier.

The VaR results tell us little about model quality, and we report that rather than dressing them as support. Filtered historical simulation recalibrates the forecast-to-quantile scaling on a trailing window, absorbing level errors, so every non-degenerate model attains near-nominal coverage (5.1–5.3%) and passes conditional coverage in 85–95% of assets. Only the random walk in realized

volatility fails, passing in 10% of assets, and it fails because its forecasts swing rather than because it sits at the wrong level. Models separate on the average VaR level at matched coverage, meaning how much capital the forecast ties up, and there the regime mixture ranks least efficient among the well-calibrated models (-0.0211 against -0.0207 for GARCH). That margin is small and we build no argument on it.

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